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Subject : DSA

Unit 3, S9, SLO1 - Queue Implementation Using Linked List

Enqueue Operation - Algorithm to insert an element VAL in a queue

Step 1: Allocate memory for the new node and name it as PTR

Step 2: SET PTR -> DATA = _____

Step 3: IF FRONT = NULL

SET FRONT = REAR = _____

SET FRONT -> NEXT = REAR -> NEXT = _____

ELSE

SET _____ = PTR

SET _____ = PTR

SET _____ = NULL

[END OF IF]

Step 4: END

Answer: Step 2: SET PTR -> DATA = VAL

Step 3: IF FRONT = NULL

SET FRONT = REAR = PTR

SET FRONT -> NEXT = REAR -> NEXT = NULL

ELSE

SET REAR -> NEXT = PTR

SET REAR = PTR

SET REAR -> NEXT = NULL

Step 4: END

Dequeue Operation – Algorithm to delete an element from a queue

Step 1: IF FRONT = NULL

Write _____

Go to Step 5

[END OF IF]

Step 2: SET PTR = _____

Step 3: SET FRONT = _____

Step 4: FREE _____

Step 5: END

Answer: Step 1: Write UNDERFLOW

Step 2: SET PTR = FRONT

Step 3: SET FRONT = FRONT -> NEXT

Step 4: FREE PTR

Step 5: END